

Reaction-Profile Matching and Maude Reachability Analysis to Prioritize Candidate Phenotype Annotations in Rare Diseases

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Abstract—Rare enzymatic diseases are often molecularly characterized but incompletely annotated in the Human Phenotype Ontology (HPO), limiting phenotype-based comparisons when clinical descriptions are sparse. We introduce a reaction-centered framework that prioritizes candidate disease–HPO associations by leveraging shared biochemical profiles. Each profile integrates six fields: causal gene, UniProt entry, EC class, Rhea reaction, cofactor, and pathway, while keeping HPO terms separate from similarity scoring. The framework combines exact biochemical-profile matching with Discrete Entity–Level Annotation–Reaction Systems (DEA–RS) models and Maude reachability analysis. Controlled validation on the phenylalanine–tyrosine pathway showed that Maude reproduced the expected physiological and perturbed outcomes, including substrate accumulation, alternative shunts, and toxic cascades. The pipeline presented in this paper was applied to 6,915 profiles from 1,276 rare diseases. It identified 117 diseases with complete data, comprising 32 HPO–unannotated diseases and 85 HPO–annotated diseases. Exact profile matching identified HPO–annotated matches for 10 of the 32 HPO–unannotated diseases, yielding 495 candidate disease–HPO associations. For selected matches, Maude tested whether enzyme and cofactor perturbations produced the expected biochemical markers. Although the strict matching criterion favors biochemical traceability over broad coverage, the framework provides traceable candidate HPO terms for expert review.

Index Terms—Reaction Systems, biochemical profile matching, Human Phenotype Ontology, Maude reachability analysis, rare enzymatic diseases, phenotype annotation prioritization

I. INTRODUCTION

For many rare enzymatic diseases, the molecular defect is known, but the clinical phenotype remains incompletely described. Sparse or missing Human Phenotype Ontology (HPO) annotations limit phenotype-based comparison [1].

Biochemical resources offer a complementary way to characterize these diseases. UniProt describes protein function [2], Rhea provides reaction records [3] and ChEBI identifies the molecular species involved [4]. These resources capture the biochemical structure of a disease, but not the effects of reduced enzyme activity, complete loss of function, or limited cofactor availability. Because kinetic parameters are rarely available for rare diseases, we bypass quantitative simulation in favor of qualitative reachability.

We adopt the qualitative Reaction Systems framework introduced in [5] and apply the Discrete Entity–Level Annotation for Reaction Systems (DEA–RS) schema. Within this framework,

a system consists of a set of entities, including enzymes, metabolites, cofactors, and pathological markers, together with a set of reactions describing their evolution. A system state is represented as a subset of all possible entities. The resulting DEA–RS rules are implemented in Maude [6], where reachability analysis is used to determine whether substrate accumulation, product reduction, or toxic marker formation can occur under specified enzyme or cofactor conditions.

The work proceeds in two stages. Stage 1 considers the phenylalanine–tyrosine Reaction System (PT–RS), namely the DEA–RS representation of the phenylalanine–tyrosine pathway developed in previous work, as a controlled baseline for the Maude encoding. The PT–RS reaction structure is translated into Maude rewriting logic and validated through reachability queries before the framework is applied to the large-scale rare–disease dataset.

Stage 2 applies the same reaction-centered framework to a large collection of rare-disease biochemical profiles. An HPO–unannotated disease is matched to an HPO–annotated disease only when the two diseases share an identical six-field biochemical profile. The HPO terms associated with the matched annotated diseases are then prioritized as candidate annotations for the unannotated disease. Selected disease pairs obtained through exact profile matching are subsequently analyzed in Maude to assess their reaction-level biochemical consistency.

The main contributions of this paper are:

- 1) a six-field reaction–profile representation that keeps biochemical comparison separate from HPO annotation;
- 2) a formal DEA–RS Maude encoding for qualitative reachability analysis under enzyme and cofactor perturbations;
- 3) an exact profile–matching pipeline that prioritizes candidate HPO terms for under-annotated enzymatic diseases;
- 4) a Maude-based biochemical consistency check that provides mechanistic context for selected candidate terms.

The proposed pipeline is not expected to replace clinical expertise, but to complement it by providing a systematic, reaction-level framework for poorly annotated enzymatic diseases.

II. RELATED WORK

The Human Phenotype Ontology provides a standardized vocabulary for describing human phenotypic abnormalities [1].

Tools built on HPO, such as Phen2Gene [7], CADA [8], and Phen2Disease [9] have made phenotype-driven gene and disease prioritization routine when clinical annotations are available. However, a key limitation is sparse or missing HPO entries, motivating work on extracting and refining annotations from clinical records [10–12].

More recent approaches extend the same phenotype-centered paradigm, including phenotypic-similarity-based variant prioritization [13], dialogue-based differential diagnosis [14], rare-disease maps [15], and deep learning for phenotype-driven disease ranking [16]. Our framework takes a different entry point: it compares diseases by their biochemical reaction profiles and consults HPO terms only after a shared enzymatic basis has been established.

A previous study formalized the phenylalanine-tyrosine pathway as a DEA-RS model, using BioReSolve for attractor analysis and dynamic negative slicing to study metabolic biomarker dependencies [5]. We reuse the same reaction structure as a controlled baseline, translate it into Maude rewriting logic, and focus on reachability properties rather than attractor analysis. The validated encoding then supports the Stage 2 reaction-profile matching and candidate HPO prioritization steps.

Consequently, this framework shifts the analysis toward formal biochemical logic and supports the prioritization of candidate HPO terms for under-documented enzymatic diseases.

III. BACKGROUND

A. HPO Annotations

The Human Phenotype Ontology (HPO) provides a standardized, hierarchically organized vocabulary for describing phenotypic abnormalities associated with human diseases [1]. Each HPO term has an identifier of the form $HP:nnnnnnnn$ and denotes a specific abnormal clinical or biological feature. An HPO annotation links a disease to one or more HPO terms, allowing disease phenotypes to be represented consistently and compared computationally.

For a disease d , we denote the set of HPO terms recorded in the Orphadata annotation file by

$$H(d) = \{h_1, h_2, \dots, h_n\}.$$

A disease is referred to as HPO-annotated when $|H(d)| > 0$ and as HPO-unannotated when $H(d) = \emptyset$. In this study, HPO-unannotated indicates the absence of recorded HPO associations in the selected data source and does not imply the absence of clinical manifestations.

B. The clinical relevance of the phenylalanine-tyrosine catabolism pathway

Phenylalanine (Phe) is an essential amino acid that must be obtained from the diet [17]. When its levels exceed protein synthesis needs, the surplus enters the phenylalanine-tyrosine catabolic pathway, which links amino-acid degradation to energy metabolism [18].

The first step in this pathway is the hydroxylation of phenylalanine to L-tyrosine (Tyr) by phenylalanine hydroxylase (PAH) [19]. This reaction requires tetrahydrobiopterin (BH_4)

as a cofactor. During catalysis, BH_4 is oxidized to BH_2 and must be regenerated by dihydropteridine reductase (DHPR) using NADH to maintain cofactor availability [20].

Excess tyrosine that is not incorporated into proteins proceeds through five successive enzymatic steps in the liver, each requiring a distinct cofactor. Tyrosine aminotransferase (TAT) converts tyrosine to 4-hydroxyphenylpyruvate and depends on pyridoxal phosphate (PLP) [21]. 4-Hydroxyphenylpyruvate dioxygenase (HPD) converts 4-hydroxyphenylpyruvate to homogentisate (HGA) and requires ferrous iron (Fe^{2+}) [22]. Homogentisate 1,2-dioxygenase (HGD) further converts homogentisate to maleylacetoacetate (MAA) and is also Fe^{2+} -dependent [23]. Maleylacetoacetate isomerase (GSTZ1) converts maleylacetoacetate to fumarylacetoacetate (FAA) using glutathione (GSH) as cofactor [24]. Finally, fumarylacetoacetate hydrolase (FAH) cleaves fumarylacetoacetate into fumarate (Fum) and acetoacetate (AcAc), a terminal step requiring Ca^{2+} and Mg^{2+} [25].

Defects in this pathway's enzymes or cofactors cause rare metabolic disorders, each presenting with a distinct biochemical and clinical profile.

PAH deficiency leads to phenylalanine accumulation, redirecting it into an alternative transamination pathway mediated by aromatic amino acid transaminase (ArAT). This alternative route produces phenylketones, including phenylpyruvate, phenylacetate, and phenyllactate, resulting in the condition known as phenylketonuria (PKU; OMIM¹ #261600) [26].

TAT impairment blocks tyrosine transamination, leading to tyrosine accumulation, which is associated with tyrosinemia type II (OMIM #276600), typically restricted to ocular and cutaneous involvement [27, 28].

HPD dysfunction leads either to 4-hydroxyphenylpyruvate (HPP) accumulation noted in tyrosinemia type III (OMIM #276710) [29] or to Hawkinsin and quinol acetate generation linked to hawkinsinuria (OMIM #140350) [30, 31], depending on enzyme activity.

HGD deficiency causes homogentisate (HGA) accumulation and can lead to benzoquinone acetate (BQA) formation, the biochemical basis of alkaptonuria (AKU; OMIM #203500) [32, 33].

GSTZ1 deficiency impairs maleylacetoacetate isomerization and is associated with maleylacetoacetate (MAA) accumulation and succinylacetoacetate (SAA) formation, a biochemical profile characteristic of maleylacetoacetate isomerase deficiency (MAAID; OMIM #617596) [34–36].

FAH deficiency blocks the terminal step of tyrosine degradation, causing hereditary tyrosinemia type I (HT-1; OMIM #276700). This triggers FAA accumulation and subsequently other toxic downstream metabolites, including succinylacetoacetate (SAA) and succinylacetone (SA). SA inhibits δ -aminolevulinic acid (ALA) dehydratase [37]. This inhibition reduces porphobilinogen (PBG) synthesis and disrupts *heme biosynthesis*², while ALA accumulation contributes to neurotoxic effects. Meanwhile, FAA and related toxic metabolites

¹OMIM stands for *Online Mendelian Inheritance in Man*, a curated catalog of human genes, genetic disorders, and traits.

²*Heme biosynthesis* is the metabolic pathway responsible for the production of heme, an essential component of hemoglobin.

drive liver failure, renal dysfunction, and hepatocellular carcinoma [35].

These conditions illustrate how defects at different points within a single metabolic pathway yield distinct chemical alterations and clinical phenotypes. Because each defect involves a different enzyme–cofactor pair and produces a distinct outcome, such as substrate accumulation, product reduction, shunt activation, or toxic marker formation, this pathway provides a controlled validation case for the DEA–RS encoding across multiple perturbation scenarios.

C. Rewriting logic and Maude

Rewriting logic is a computational framework in which system states are terms in an equational theory and transitions are described by rewrite rules [38, 39]. Maude implements this framework and supports simulation, search, reachability analysis, and model checking [40–42] <https://maude.lcc.um.es/maude-manual/>. Reaction systems can be represented by mapping entities to terms and reactions to rewrite rules, making it possible to study which states are reachable from a given initial configuration [43].

In this work, Maude reachability queries verify whether specific biochemical markers arise under normal, hypomorphic, null, or cofactor–limited conditions for each enzymatic step. This provides a qualitative consistency check for the reaction–level assumptions used later in the HPO prioritization framework.

IV. MATERIALS AND METHODS

The workflow proposed in this study is composed of two stages (Figure 1). Stage 1 validates the Maude encoding against the PT–RS baseline, a curated phenylalanine–tyrosine pathway with eight enzymatic reactions and a controlled set of reachability properties. Stage 2 applies the same modeling logic to a rare-disease dataset, where diseases are compared by exact biochemical reaction profiles and selected target–donor pairs are checked in Maude. Stage 1 thus confirms that the DEA–RS encoding reproduces known biochemical outcomes before the same rule structure and reachability-query scheme are applied to the selected reaction–level models used in Stage 2.

A. Reaction Systems and Interactive Processes

Reaction Systems (RSs) were introduced by Ehrenfeucht and Rozenberg [44]. They provide a qualitative framework for representing biochemical dynamics in terms of the presence or absence of molecular entities, without relying on concentrations or kinetic rates [45].

A Reaction System is a pair

$$\mathcal{A} = (S, A),$$

where S is a finite set of entities and A is a finite set of reactions over S . In a biochemical model, the entities may represent metabolites, enzymes, cofactors, enzyme activity states, or pathological markers.

A reaction $a \in A$ is represented by a triple

$$a = (R, I, P),$$

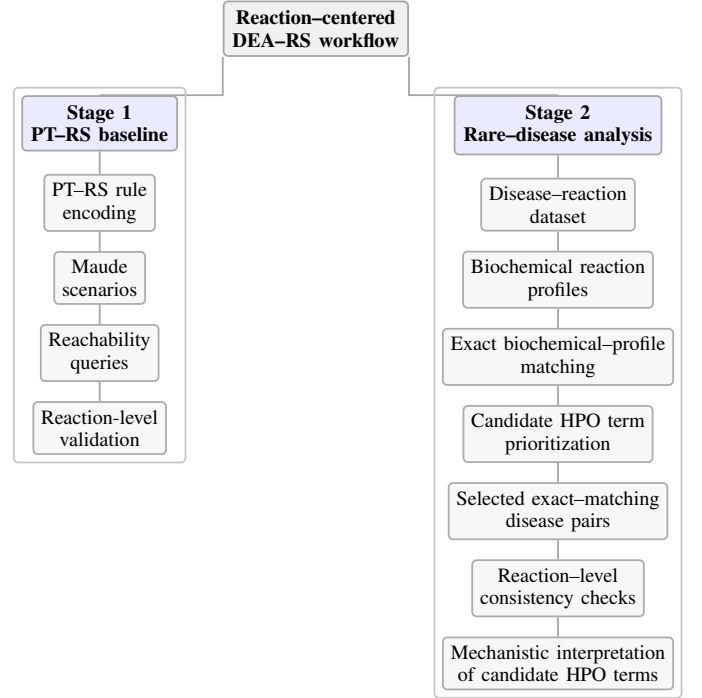


Fig. 1: Overview of the two-stage workflow. Stage 1 uses the PT–RS phenylalanine–tyrosine pathway as a controlled baseline for Maude reachability validation. Stage 2 applies exact biochemical–profile matching to rare enzymatic diseases, prioritizes candidate HPO terms from exact profile matches, and evaluates selected exact–matching disease pairs for reaction–level biochemical consistency.

where $R \subseteq S$ is the non-empty set of reactants required for the reaction, $I \subseteq S$ is the set of inhibitors, and $P \subseteq S$ is the non-empty set of products, with $R \cap I = \emptyset$. In our models, I is allowed to be empty in order to represent reactions that have no explicit inhibitor.

Given a state $W \subseteq S$, reaction a is enabled when all its reactants are present and none of its inhibitors is present:

$$\text{en}_a(W) \equiv (R \subseteq W) \wedge (I \cap W = \emptyset).$$

The result of a on W is therefore

$$\text{res}_a(W) = \begin{cases} P, & \text{if } \text{en}_a(W), \\ \emptyset, & \text{otherwise.} \end{cases}$$

All reactions are evaluated with respect to the same current state. The result produced by the complete reaction set A is

$$\text{res}_A(W) = \bigcup_{a \in A} \text{res}_a(W).$$

The formalism is qualitative: entities are either present or absent, their multiplicities are not considered, and an entity persists only if it is reproduced by an enabled reaction or supplied again by the environment.

Because biochemical systems interact with their environment, their evolution is described through interactive processes. Let $\mathcal{A} = (S, A)$ be a Reaction System. An interactive process is a pair

$$\pi = (\gamma, \delta),$$

where

$$\gamma = \{C_i\}_{i=0}^n \quad \text{and} \quad \delta = \{D_i\}_{i=0}^n$$

are, respectively, the context sequence and the result sequence.

For each evolution step, $C_i \subseteq S$ contains the entities supplied by the environment, whereas $D_i \subseteq S$ contains the entities generated by the Reaction System. Starting from $D_0 = \emptyset$, the evolution is defined by

$$D_{i+1} = \text{res}_A(D_i \cup C_i), \quad i = 0, \dots, n-1.$$

The complete state at step i is

$$W_i = C_i \cup D_i,$$

and

$$\tau = (W_0, \dots, W_n)$$

is the corresponding state sequence. Thus, the context sequence γ represents the interaction with the environment, while the result sequence δ is determined by γ and the reaction set A . In our biochemical scenarios, the contexts specify the available substrates, cofactors, and enzyme activity conditions, including the perturbations to be analyzed.

B. Stage 1: Validation Against the PT-RS Baseline

Stage 1 evaluates our Maude implementation using the phenylalanine–tyrosine (Phe–Tyr) catabolism pathway as a controlled validation case. The reference model is the curated Phe–Tyr Reaction System (PT-RS) developed in previous work [5]. Its DEA-RS rules are translated into Maude rewriting logic, and reachability queries are used to analyze physiological and perturbed contexts.

The analysis verifies whether the expected pathway products are reachable under normal enzyme and cofactor conditions and whether substrate accumulation, reduced product formation, alternative shunts, and pathological markers become reachable under the corresponding perturbations. After validating the Maude encoding against the expected PT-RS behavior, the same modeling and analysis procedure is applied to the selected rare-disease profiles.

1) *The Reference Pathway Reactions*: The reference pathway, reported in Table I, comprises eight enzymatic reactions. Seven reactions describe the physiological flux from phenylalanine to fumarate and acetoacetate. The eighth is the ArAT-catalyzed transamination of phenylalanine to phenylpyruvate, a minor alternative shunt that becomes relevant when phenylalanine hydroxylase (PAH) activity is impaired.

2) *DEA-RS Formalization*: Building on the Reaction Systems semantics introduced above, we apply the Discrete Entity-Level Annotation for Reaction Systems (DEA-RS) schema to the Phe–Tyr pathway. The DEA-RS schema preserves the standard (R, I, P) reaction structure while enriching the entity set with discrete annotations that represent physiological and pathological biochemical conditions. In the resulting model, the finite entity set S includes metabolites, cofactors, functional enzymes, altered enzyme activity states, and pathological markers.

Enzyme dysfunction is represented through the suffix `-hypo` for reduced activity and the suffix `-null` for absent activity.

These annotated entities inhibit the corresponding physiological rules and enable the associated perturbed rules. Similarly, abnormal metabolite states are represented through `-acc` for accumulation and `-low` for reduced availability or production. These annotations make the biochemical interpretation of each state explicit while retaining the qualitative presence–absence semantics of Reaction Systems.

The context C_i specifies the scenario-dependent entities supplied at evolution step i , such as substrates, cofactors, and the selected functional or perturbed enzyme state. The result set D_i contains the entities produced by the enabled DEA-RS rules, and $W_i = C_i \cup D_i$ is the complete state represented in Maude. Under this correspondence, the Maude rewriting rules implement the transition

$$D_{i+1} = \text{res}_A(W_i),$$

and reachability analysis determines whether a target metabolite or pathological marker occurs in at least one reachable state. In particular, for a target entity $m \in S$, the analysis tests whether there exists a reachable state W_j such that $m \in W_j$.

For example, the physiological PAH reaction, corresponding to rule M2 in Table XVII, is represented as

$$a_{\text{PAH}} = (\{\text{Phe}, \text{BH}_4, \text{PAH-Fe}^{2+}, \text{O}_2\}, \\ \{\text{PAH-hypo}, \text{PAH-null}\}, \\ \{\text{Tyr}, \text{BH}_2, \text{H}_2\text{O}, \text{PAH-Fe}^{2+}\}).$$

This rule is enabled when phenylalanine, tetrahydrobiopterin, oxygen, and functional PAH with its iron cofactor are present, while neither the reduced-activity nor the null PAH state is present. Its products are then generated in the next result set.

The Maude specification follows the (R, I, P) structure. Table XVII reports the 38 physiological and pathological DEA-RS rules used in our validation model. These rules cover the seven physiological enzymatic steps, the ArAT-mediated phenylalanine shunt, the altered enzyme and cofactor conditions, and the resulting accumulation, reduction, and toxic-marker rules. Compared with the original 39-rule PT-RS model, our Maude validation model omits the reverse TAT reaction.

3) *Physiological and Perturbed Scenarios*: The initial conditions define the biological setting explored through reachability analysis. For each scenario s , we define an interactive process

$$\pi_s = (\gamma_s, \delta_s),$$

where

$$\gamma_s = \{C_i^s\}_{i=0}^n \quad \text{and} \quad \delta_s = \{D_i^s\}_{i=0}^n$$

are, respectively, the context sequence and the result sequence. At each evolution step, $C_i^s \subseteq S$ contains the entities supplied by the environment, whereas $D_i^s \subseteq S$ contains the entities generated by the enabled reactions. Starting from $D_0^s = \emptyset$, the complete state is

$$W_i^s = C_i^s \cup D_i^s,$$

and the result generated for the next step is

$$D_{i+1}^s = \text{res}_A(W_i^s).$$

Each scenario is represented by a corresponding input process

$$\mathcal{I}_s = \{\text{Phe}, \text{BH}_4\}.\mathbf{0} \parallel \text{rec } X.\text{Env}_s.X.$$

TABLE I: Reference enzymatic reaction set used for the Phe–Tyr validation case.

Rhea ID	Biochemical reaction	EC	Enzyme name	Cofactor
RHEA:32251	6,7-dihydrobiopterin + NADH + H ⁺ → 5,6,7,8-tetrahydrobiopterin + NAD ⁺	1.5.1.34	DHPR: 6,7-dihydropteridine reductase	ND
RHEA:20273	L-phenylalanine + 5,6,7,8-tetrahydrobiopterin + O ₂ → L-tyrosine + 6,7-dihydrobiopterin + H ₂ O	1.14.16.1	PAH: phenylalanine 4-monooxygenase	Fe ²⁺ /BH ₄
RHEA:25152	L-phenylalanine + 2-oxoglutarate → phenylpyruvate + L-glutamate	2.6.1.57; 2.6.1.5	ArAT: aromatic-amino-acid transaminase; TAT: tyrosine transaminase	PLP
RHEA:15093	L-tyrosine + 2-oxoglutarate → 3-(4-hydroxyphenyl)pyruvate + L-glutamate	2.6.1.5	TAT: tyrosine transaminase	PLP
RHEA:16189	3-(4-hydroxyphenyl)pyruvate + O ₂ → homogentisate + CO ₂	1.13.11.27	HPD: 4-hydroxyphenylpyruvate dioxygenase	Fe ²⁺
RHEA:15449	Homogentisate + O ₂ → 4-maleylacetoacetate + H ⁺	1.13.11.5	HGD: homogentisate 1,2-dioxygenase	Fe ²⁺
RHEA:14817	4-maleylacetoacetate → fumarylacetoacetate	5.2.1.2	GSTZ1: maleylacetoacetate isomerase	GSH
RHEA:10244	4-fumarylacetoacetate + H ₂ O → acetoacetate + fumarate + H ⁺	3.7.1.2	FAH: fumarylacetoacetase	Ca ²⁺ /Mg ²⁺

Key: Rhea ID = identifier in the Rhea reaction database; EC = Enzyme Commission number; ND = not defined. *Enzymes*: DHPR = 6,7-dihydropteridine reductase; PAH = phenylalanine hydroxylase; ArAT = aromatic-amino-acid transaminase; TAT = tyrosine aminotransferase; HPD = 4-hydroxyphenylpyruvate dioxygenase; HGD = homogentisate 1,2-dioxygenase; GSTZ1 = glutathione S-transferase zeta 1, maleylacetoacetate isomerase; FAH = fumarylacetoacetate hydrolase. *Metabolites and cofactors*: BH₂ = dihydrobiopterin; BH₄ = tetrahydrobiopterin; Phe = phenylalanine; Tyr = tyrosine; HPP = 4-hydroxyphenylpyruvate; HGA = homogentisate; MAA = maleylacetoacetate; FAA = fumarylacetoacetate; AcAc = acetoacetate; Fum = fumarate; PPy = phenylpyruvate; Glu = glutamate; PLP = pyridoxal 5'-phosphate; α-KG = α-ketoglutarate; GSH = glutathione.

TABLE II: Physiological and pathological DEA–RS reaction rules in Maude

ID	Step / condition	Reactants, <i>R</i>	Inhibitors, <i>I</i>	Products, <i>P</i>
M1	DHPR BH4 regeneration	BH2, NADH, H, DHPR	DHPR-hypo, DHPR-null	BH4, NAD, DHPR
M2	PAH Phe hydroxylation	Phe, BH4, PAH-Fe2, O2	PAH-hypo, PAH-null	Tyr, BH2, H2O, PAH-Fe2
M3	PAH-null Phe accumulation	Phe, BH4, O2, PAH-null	PAH-Fe2	Phe-acc, BH4, O2, PAH-null
M4	PAH-null shunt	Phe-acc, alpha-KG, PLP, ArAT, PAH-null	PAH-Fe2	PPy, Glu, PLP, Phe-acc, ArAT, PAH-null
M5	PAH-hypo activity	Phe, BH4, PAH-hypo, O2	PAH-Fe2	Tyr-low, BH2-low, H2O, Phe-acc, PAH-hypo
M6	PAH-hypo shunt	Phe-acc, alpha-KG, PLP, ArAT-hypo, PAH-hypo	PAH-Fe2	PPy, Glu, PLP, Phe-acc, ArAT-hypo, PAH-hypo
M7	DHPR-hypo	BH2, NADH, H, DHPR-hypo	DHPR	BH4-low, NAD, DHPR-hypo
M8	DHPR-hypo BH4-limited PAH	Phe, BH4-low, O2, PAH-Fe2	PAH-hypo, PAH-null, BH4	Tyr-low, BH2-low, H2O, Phe-acc, PAH-Fe2
M9	DHPR-hypo	BH2-low, NADH, H, DHPR-hypo	DHPR	BH4-low, NAD, DHPR-hypo
M10	DHPR-hypo	Phe-acc, BH4-low, O2, PAH-Fe2	PAH-hypo, PAH-null	Tyr-low, BH2-low, H2O, Phe-acc, PAH-Fe2
M11	DHPR-null	BH2, NADH, H, DHPR-null	DHPR	BH2-acc, NADH, H, DHPR-null
M12	TAT	Tyr, PLP, alpha-KG, TAT	TAT-null, TAT-low	HPP, Glu, PLP, TAT
M13	TAT-low	Tyr, PLP, alpha-KG, TAT-low	TAT	HPP-low, Glu-low, Tyr-acc, PLP, TAT-low
M14	TAT-null	Tyr, PLP, alpha-KG, TAT-null	TAT	Tyr-acc, alpha-KG, PLP, TAT-null
M15	PLP deficiency	Tyr, PLP-low, alpha-KG, TAT	PLP	HPP-low, Glu-low, Tyr-acc, PLP-low, TAT
M16	HPD	HPP, O2, HPD-Fe2	HPD-null, HPD-hypo	HGA, CO2, HPD-Fe2
M17	HPD-hypo	HPP, O2, HPD-hypo	HPD-Fe2	HGA-low, CO2, QA, HPP-acc, HPD-hypo
M18	Hawkinsin marker	QA, GSH	∅	Hawkinsin-acc
M19	HPD-null	HPP, O2, HPD-null	HPD-Fe2	HPP-acc, HPP, O2, HPD-null
M20	HGD	HGA, O2, HGD-Fe2	HGD-null, HGD-hypo	MAA, H, HGD-Fe2
M21	HGD-hypo	HGA, O2, HGD-hypo	HGD-Fe2	MAA-low, H, HGA-acc, HGD-hypo
M22	HGD-null	HGA, O2, HGD-null	HGD-Fe2	HGA-acc, O2, HGD-null
M23	Alkaptonuria marker	HGA-acc, O2	HGD-Fe2	BQA, HGA-acc, O2
M24	GSTZ1	MAA, GSH, GSTZ1	GSTZ1-null, GSTZ1-hypo	FAA, GSH, GSTZ1
M25	GSTZ1-hypo	MAA, GSH, GSTZ1-hypo	GSTZ1	FAA-low, MAA-acc, GSH, GSTZ1-hypo
M26	GSTZ1-null	MAA, GSH, GSTZ1-null	GSTZ1	MAA-acc, GSH, GSTZ1-null
M27	SAA marker	MAA-acc, O2	∅	SAA, CO2, MAA-acc
M28	FAH	FAA, H2O, FAH-Mg2-Ca2	FAH-null, FAH-hypo	AcAc, Fum, H, FAH-Mg2-Ca2
M29	FAH-hypo	FAA, H2O, FAH-hypo	FAH-Mg2-Ca2	AcAc-low, Fum-low, FAA-acc, H, FAH-hypo
M30	FAH-hypo	FAA-acc, FAH-hypo	FAH-Mg2-Ca2	SAA, CO2, FAA-acc, FAH-hypo
M31	SAA cascade, FAH-hypo	SAA	∅	SA, SAA
M32	SAA cascade, FAH-hypo	ALA, SA	∅	ALA-acc, SA
M33	SAA cascade, FAH-hypo	ALA-acc, SA	∅	PBG-low, SA
M34	FAH-null	FAA, H2O, FAH-null	FAH-Mg2-Ca2	FAA-acc, H2O, FAH-null
M35	FAH-null	FAA-acc, FAH-null	∅	SAA, FAA-acc, FAH-null
M36	SAA cascade, FAH-null	SAA	∅	SA, SAA
M37	SAA cascade, FAH-null	ALA, SA	∅	ALA-acc, SA
M38	SAA cascade, FAH-null	ALA-acc, SA	∅	PBG-low, SA

Key: ∅ = empty inhibitor set, M1–M38 denote the Maude reaction rules.

In this notation, a prefix $C.P$ supplies the context C and then continues as the process P , 0 denotes termination, $\parallel$ denotes parallel composition, and $\text{rec } X.P$ denotes the recursive repetition of the process P .

The first component supplies the initial context $\{\text{Phe}, \text{BH}_4\}$ only once, whereas the recursive component supplies the scenario-specific environment $\text{Env}_s \subseteq S$ at every evolution

step. The resulting context sequence is

$$C_0^s = \{\text{Phe}, \text{BH}_4\} \cup \text{Env}_s, \quad C_i^s = \text{Env}_s \quad \text{for } i \geq 1.$$

The baseline environment shared by all scenarios is

$$\text{Env}_{\text{base}} = \{\text{O}_2, \text{H}_2\text{O}, \text{NADH}, \text{H}, \text{GSH}, \text{ALA}\}.$$

The normal environment extends this set with the functional enzymes, cofactors, and additional metabolites required by the pathway:

$$\text{Env}_{\text{norm}} = \text{Env}_{\text{base}} \cup \{\alpha\text{-KG}, \text{PLP}, \text{PAH-Fe}^{2+}, \text{DHPR}, \text{ArAT}, \text{TAT}, \text{HPD-Fe}^{2+}, \text{HGD-Fe}^{2+}, \text{GSTZ1}, \text{FAH-Mg}^{2+}\text{-Ca}^{2+}\}.$$

The corresponding normal input process is

$$\mathcal{I}_{\text{norm}} = \{\text{Phe}, \text{BH}_4\}.\mathbf{0} \parallel \text{rec } X.\text{Env}_{\text{norm}}.X.$$

A perturbed scenario is obtained by changing one component of the recurring environment. For an enzyme E , we first define the environment in which the functional enzyme is removed:

$$\text{Env}_{\text{no-}E} = \text{Env}_{\text{norm}} \setminus \{E\}.$$

The complete perturbed environment is then obtained by introducing either the reduced-function or the null variant $E^* \in \{E\text{-hypo}, E\text{-null}\}$:

$$\text{Env}_{E^*} = \text{Env}_{\text{no-}E} \cup \{E^*\}.$$

The corresponding input process is

$$\mathcal{I}_{E^*} = \{\text{Phe}, \text{BH}_4\}.\mathbf{0} \parallel \text{rec } X.\text{Env}_{E^*}.X.$$

Cofactor limitation is modeled analogously. In the PLP-limited scenario, PLP is replaced by PLP-low, enabling the rule that generates reduced HPP and glutamate together with tyrosine accumulation.

All scenarios share the same initial context and baseline environment and differ in only one enzyme or cofactor state. Differences in reachable accumulations, reduced products, alternative metabolites, or pathological markers can therefore be associated, within the encoded model, with the selected perturbation. The scenario environments are summarized in Table XVIII.

4) *Maude Reachability Analysis*: The Maude model is used to determine whether a biochemical marker is reachable from the input process associated with a given scenario. For each scenario-specific input process $\mathcal{I}_s$ and target marker m , we use a query of the following form:

```
search [1] init( $\mathcal{I}_s$ ) =>* STATE such that STATE |= ( $m$ )
```

Here, `search [1]` asks Maude to return one state satisfying the target condition, `=>*` denotes zero or more rewrite steps, and `STATE |=  $m$` checks whether marker m is present in the reached state. A returned state confirms reachability, whereas when the search terminates, `No solution` indicates that the marker is not reachable within the encoded model. Individual markers are checked through separate reachability queries. When a property concerns the simultaneous presence of two markers in the same state, their conditions are combined in a single reachability query. For multi-step pathways, intermediate and downstream markers are tested separately to determine

whether an enzyme or cofactor perturbation introduced earlier in the pathway produces the expected downstream effects.

The validation properties reported in Table IV are grouped into four classes. Enzyme activity-state exclusivity (P1–P2) checks that incompatible states of the same enzyme, such as functional, hypomorphic, and null states, do not occur simultaneously in a reachable state. Physiological integrity (P3, P6, and P9) verifies that pathological markers or incompatible biochemical conditions are not reachable under normal conditions. Disease-specific marker reachability (P4–P5 and P7–P8) checks whether the expected biochemical markers become reachable under the corresponding enzyme perturbations. Pathway-level outcome reachability (P10–P11) evaluates whether alternative-shunt products or terminal pathway metabolites are reachable under the appropriate scenarios.

Properties involving a single marker are evaluated through individual reachability queries. When a property concerns the simultaneous presence of two markers in the same state, their conditions are combined in a single query.

To illustrate how the properties in Table IV are evaluated, consider P7, which tests whether HGA accumulation is reachable under reduced HGD activity:

```
Maude> search [1] init(input-process-HGD-hypo) =>*
STATE such that STATE |= (HGA-acc) .
```

This query asks Maude to find a state reachable from the `HGD-hypo` scenario that contains `HGA-acc`. A returned state therefore confirms the reachability of HGA accumulation under reduced HGD activity. The downstream marker `BQA` is evaluated through a separate query.

As an example of a negative reachability property, P6 tests whether `Hawkinsin-acc` is reachable under normal conditions:

```
Maude> search [1] init(input-process-normal) =>*
STATE such that STATE |= (Hawkinsin-acc) .
```

For this property, the expected outcome is `No solution`, indicating that no reachable state contains `Hawkinsin-acc` under the normal scenario. The precursor marker `QA` is checked through a separate query. The outcomes of these and the remaining reachability queries are reported in the Results section.

C. Stage 2: Rare-Disease Dataset Construction and Candidate HPO Prioritization

Stage 2 extends the validated DEA-RS/Maude framework from the controlled PT-RS pathway to a multi-source dataset of rare enzymatic diseases. Biochemical and HPO data are extracted, harmonized, and filtered to construct complete reaction profiles for exact disease matching and subsequent reaction-level analysis in Maude.

The workflow has two complementary tasks. First, diseases without recorded HPO annotations are compared with HPO-annotated diseases using exact six-field reaction-profile matching. HPO terms associated with matching annotated diseases are then prioritized as candidates for the unannotated diseases. Second, selected matched disease pairs are encoded as reaction-specific DEA-RS models and analyzed in Maude

TABLE III: Maude environments and environment templates used to construct the physiological and perturbed pathway scenarios.

Scenario and environment	Environment entities	Biological interpretation
Normal physiological scenario env-normal	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, TAT, HPD-Fe2, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	Reference physiological scenario with functional enzymes and cofactors.
PAH deficiency env-no-PAH	env-base, alpha-KG, PLP, DHPR, ArAT, TAT, HPD-Fe2, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	PKU-like scenario in which functional PAH is replaced by PAH-hypo or PAH-null.
DHPR deficiency env-no-DHPR	env-base, alpha-KG, PLP, PAH-Fe2, ArAT, TAT, HPD-Fe2, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	DHPR-deficient scenario in which functional DHPR is replaced by DHPR-hypo or DHPR-null, impairing BH ₄ regeneration.
TAT deficiency env-no-TAT	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, HPD-Fe2, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	Tyrosinemia type II-like scenario in which functional TAT is replaced by TAT-low or TAT-null.
PLP limitation env-PLP-low	env-base, alpha-KG, PLP-low, PAH-Fe2, DHPR, ArAT, TAT, HPD-Fe2, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	PLP-limited scenario in which normal PLP is replaced by PLP-low, reducing cofactor support for TAT-mediated transamination.
HPD deficiency env-no-HPD	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, TAT, HGD-Fe2, GSTZ1, FAH-Mg2-Ca2	HPD-deficient scenario in which functional HPD is replaced by HPD-hypo or HPD-null. The two variants represent Hawkinsinuria-like and tyrosinemia type III-like conditions, respectively.
HGD deficiency env-no-HGD	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, TAT, HPD-Fe2, GSTZ1, FAH-Mg2-Ca2	Alkaptonuria-like scenario in which functional HGD is replaced by HGD-hypo or HGD-null.
GSTZ1 deficiency env-no-GSTZ1	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, TAT, HPD-Fe2, HGD-Fe2, FAH-Mg2-Ca2	GSTZ1-deficient scenario in which functional GSTZ1 is replaced by GSTZ1-hypo or GSTZ1-null.
FAH deficiency env-no-FAH	env-base, alpha-KG, PLP, PAH-Fe2, DHPR, ArAT, TAT, HPD-Fe2, HGD-Fe2, GSTZ1	Tyrosinemia type I-like scenario in which functional FAH is replaced by FAH-hypo or FAH-null.

Key: env-base supplies O₂, H₂O, NADH, H, GSH, ALA in all scenarios. Each env-no-E denotes an environment template in which the functional enzyme E is removed. The complete perturbed environment is obtained by adding either the corresponding -hypo or -null variant. In the PLP-limited scenario, PLP is replaced directly by PLP-low. All input processes share the initial component $\backslash\{\text{Phe}, \text{BH}_4\}.0$.

TABLE IV: PT-RS validation properties encoded as Maude reachability queries.

ID	Validation class	Scenario	Query target and expected outcome
P1	Enzyme activity-state exclusivity	Normal context	PAH-null is unreachable and does not co-occur with PAH-Fe2.
P2	Enzyme activity-state exclusivity	Normal context	No mutually incompatible activity states of the same enzyme are reachable simultaneously.
P3	Physiological integrity	Normal context	Phe-acc is unreachable.
P4	Disease-specific marker reachability	PAH-null context	Phe-acc and PPy are reachable in the same state.
P5	Disease-specific marker reachability	HPD-hypo context	QA and Hawkinsin-acc are reachable.
P6	Physiological integrity	Normal context	QA and Hawkinsin-acc are unreachable.
P7	Disease-specific marker reachability	HGD-hypo or HGD-null context	HGA-acc and BQA are reachable.
P8	Disease-specific marker reachability	FAH-hypo or FAH-null context	FAA-acc, SAA, SA, ALA-acc, and PBG-low are reachable.
P9	Physiological integrity	Normal context	Phe-acc and Tyr-acc are not reachable simultaneously.
P10	Pathway-level outcome reachability	PAH-null context	PPy and Glu are reachable through the ArAT-mediated transamination shunt.
P11	Pathway-level outcome reachability	Normal context	The terminal products AcAc and Fum are reachable.

to test whether the expected markers are reachable under the corresponding enzyme or cofactor perturbation. Reaction-profile matching, therefore, supports large-scale candidate prioritization, while Maude verifies the reaction-level consistency of selected matches.

Disease information was integrated from Orphanet³, UniProt [2], Rhea [3], and ChEBI [4]. Disorder names and ORPHA codes were extracted from en_product1.xml, HPO annotations from en_product4.xml, and disease-

gene associations from en_product6.xml.

Only genes associated with metabolic or enzymatic disorders were retained. Each gene was mapped to a reviewed human UniProt entry to retrieve its identifier, EC classification, pathway, and cofactor information from UniProt records and the UniProt API. Enzymes were then linked to their physiological reactions in Rhea, and the participating chemical entities were normalized using ChEBI identifiers. The integrated data were organized into six-field reaction profiles containing the causal gene, UniProt entry, EC class, Rhea reaction, cofactor, and pathway. Diseases lacking cofactor or pathway information

³<https://sciences.orphadata.com/>

were excluded before profile construction, and redundant records were collapsed.

Each disease d was represented by a reaction–profile set $\mathcal{P}(d)$ and by the HPO annotation set $H(d)$ defined in Section III-A. The latter was used to partition the diseases into HPO-annotated and HPO-unannotated sets and to collect candidate terms after exact biochemical-profile matching. It was not included in the matching criterion.

Figure 2 summarizes the data construction, profile matching, candidate prioritization, and Maude verification pipeline.

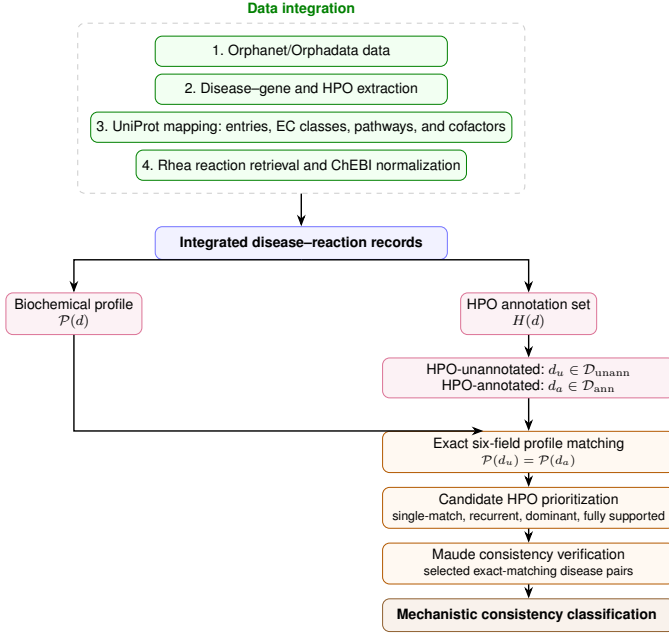


Fig. 2: Stage 2 workflow for data integration, exact biochemical-profile matching, candidate HPO prioritization, and Maude consistency verification.

The complete dataset and associated analysis files are available in the [project GitHub repository](#) to support reproducibility and open science.

1) *Disease Biochemical Profiles*: After data integration and filtering, six fields were used for profile matching: causal gene, reviewed human UniProt entry, EC class, Rhea reaction identifier, cofactor, and metabolic pathway. The remaining dataset fields were not included in the matching criterion.

For each disease–reaction association j of disease d , these fields define the six–field descriptor

$$\Phi_j(d) = (g_j, u_j, e_j, r_j, c_j, \lambda_j),$$

where g_j is the causal gene, u_j the UniProt entry, e_j the EC class, r_j the Rhea reaction identifier, c_j the cofactor set, and λ_j the associated metabolic pathway set.

Because a disease may be associated with several reactions, its complete biochemical profile is

$$\mathcal{P}(d) = \{\Phi_1(d), \Phi_2(d), \dots, \Phi_m(d)\},$$

where m is the number of disease–reaction associations for disease d . For example, let d_{69723} denote Tyrosinemia type 3 (ORPHA: 69723)⁴. Its disease–reaction descriptor is

$$\Phi_1(d_{69723}) = (\text{HPD}, \text{P32754}, 1.13.11.27, \text{RHEA:16189}, \{\text{Fe}\}, \lambda_{\text{HPD}}).$$

The six entries correspond, in order, to g_1 , u_1 , e_1 , r_1 , c_1 , and λ_1 in the general definition of $\Phi_j(d)$. Here, Fe denotes the iron cofactor, and λ_{HPD} denotes the associated metabolic pathway set.⁵ For this matched profile, the complete biochemical profile contains one descriptor:

$$\mathcal{P}(d_{69723}) = \{\Phi_1(d_{69723})\}.$$

HPO annotations are represented separately as

$$H(d) = \{h_1, h_2, \dots, h_n\},$$

where $|H(d)|$ is the number of HPO terms associated with disease d .

2) *Exact Profile Matching*: Let $\mathcal{D}$ denote the set of analyzed diseases, partitioned into

$$\mathcal{D}_{\text{unann}} = \{d \in \mathcal{D} : H(d) = \emptyset\}$$

and

$$\mathcal{D}_{\text{ann}} = \{d \in \mathcal{D} : |H(d)| > 0\}.$$

Each disease $d_u \in \mathcal{D}_{\text{unann}}$ is compared with every disease $d_a \in \mathcal{D}_{\text{ann}}$ using

$$\text{Sim}_{\text{ex}}(d_u, d_a) = \begin{cases} 1, & \text{if } \mathcal{P}(d_u) = \mathcal{P}(d_a), \\ 0, & \text{otherwise.} \end{cases}$$

A value of 1 indicates identity between the complete six–field biochemical profiles; partial matches are excluded. For example, Tyrosinemia type 3 (d_{69723}) and Hawkinsinuria (d_{2118}) share the same descriptor:

$$\Phi_1(d_{69723}) = \Phi_1(d_{2118}),$$

implying

$$\text{Sim}_{\text{ex}}(d_{69723}, d_{2118}) = 1.$$

However, their HPO annotation sets are different:

$$H(d_{69723}) = \emptyset, \quad |H(d_{2118})| = 9.$$

This exact match allows the nine terms in $H(d_{2118})$ to be considered as candidate HPO annotations for d_{69723} .

3) *Filter–Sensitivity Analysis*: To determine whether the cofactor/pathway filter changed the matching results, the exact–matching pipeline was repeated under two settings. The first applied the filter, requiring at least one annotated cofactor and one metabolic pathway, whereas the second omitted it. In both settings, matching was restricted to complete six–field profiles. The outputs were compared using the numbers of analyzable diseases, HPO–unannotated and HPO–annotated diseases, exact matches, candidate HPO associations, and candidate–support categories.

⁴ORPHA codes are numerical identifiers assigned by Orphanet to rare diseases.

⁵The pathway annotations are *acetoacetate* and *fumarate* from *L-phenylalanine: step 3/6, amino-acid degradation*, and *L-phenylalanine degradation*.

TABLE V: Mechanistic-consistency classes used to interpret candidate HPO associations analyzed in Maude.

Class	Mechanistic criterion	Interpretation
Strong	The Maude analysis supports a direct biochemical link between the reaction-level outcome and the candidate HPO term.	Direct mechanistic support.
Moderate	The verified reaction-level outcome is indirectly or downstream related to the candidate HPO term.	Plausible but indirect support.
Context-dependent	The reaction-level outcome is consistent with the candidate term, but its interpretation requires additional biological or clinical context.	Support depends on disease context.
Low	The verified reaction-level outcome provides no clear mechanistic link to the candidate HPO term.	Limited mechanistic support.

4) *Internal Baseline Comparison*: To assess how restrictive the six-field criterion was, we compared it with seven simpler configurations: Gene-only, UniProt Entry-only, EC-only, Rhea-only, Pathway-only, Gene+EC+Rhea, and Gene+Entry+EC+Rhea.

For each configuration, matching was performed using only the selected fields, and candidate HPO terms were prioritized using the same procedure as in the full analysis. We compared the number of HPO-unannotated diseases with at least one match and the mean number of candidate HPO associations per matched disease. The first measure reflects how many diseases were matched, whereas the second reflects the average number of candidates generated for each matched disease.

5) *HPO Candidate Prioritization*: For each disease $d_u \in \mathcal{D}_{\text{unann}}$, the set of exact-matching HPO-annotated diseases is

$$\mathcal{M}(d_u) = \{d_a \in \mathcal{D}_{\text{ann}} : \text{Sim}_{\text{ex}}(d_u, d_a) = 1\}.$$

The candidate HPO terms are collected from these diseases:

$$H_{\text{cand}}(d_u) = \bigcup_{d_a \in \mathcal{M}(d_u)} H(d_a).$$

For each candidate term h , its support count is

$$s_h(d_u) = |\{d_a \in \mathcal{M}(d_u) : h \in H(d_a)\}|,$$

that is, the number of exact-matching HPO-annotated diseases containing h . The single-match and recurrent candidate sets are defined as

$$H_{\text{single}}(d_u) = \{h \in H_{\text{cand}}(d_u) : s_h(d_u) = 1\},$$

$$H_{\text{rec}}(d_u) = \{h \in H_{\text{cand}}(d_u) : s_h(d_u) \geq 2\}.$$

Among the recurrent terms, the dominant candidates are those with the highest support, where q ranges over the recurrent candidate terms:

$$H_{\text{dom}}(d_u) = \left\{ h \in H_{\text{rec}}(d_u) : s_h(d_u) = \max_{q \in H_{\text{rec}}(d_u)} s_q(d_u) \right\}.$$

If $H_{\text{rec}}(d_u) = \emptyset$, then $H_{\text{dom}}(d_u) = \emptyset$. A dominant candidate is fully supported when it occurs in every exact-matching HPO-annotated disease:

$$H_{\text{full}}(d_u) = \{h \in H_{\text{dom}}(d_u) : s_h(d_u) = |\mathcal{M}(d_u)|\}.$$

The support sets satisfy

$$H_{\text{full}}(d_u) \subseteq H_{\text{dom}}(d_u) \subseteq H_{\text{rec}}(d_u) \subseteq H_{\text{cand}}(d_u),$$

whereas

$$H_{\text{single}}(d_u) \cap H_{\text{rec}}(d_u) = \emptyset.$$

TABLE VI: Reporting categories for candidate HPO terms based on support among exact-matching HPO-annotated diseases.

Candidate type	Reporting condition	Interpretation
Single-match (H_{single})	$h \in H_{\text{single}}(d_u)$	Supported by one exact-matching disease.
Recurrent (H_{rec})	$h \in H_{\text{rec}}(d_u) \setminus H_{\text{dom}}(d_u)$	Supported by at least two exact-matching diseases.
Dominant (H_{dom})	$h \in H_{\text{dom}}(d_u) \setminus H_{\text{full}}(d_u)$	Recurrent term with the highest support.
Fully supported (H_{full})	$h \in H_{\text{full}}(d_u)$	Present in every exact-matching HPO-annotated disease.

Algorithm 1 Exact biochemical-profile matching and HPO prioritization

Require: Integrated disease-reaction records

Ensure: Candidate HPO terms and support categories

- 1: Normalize the six profile fields and exclude incomplete descriptors
- 2: Build $\mathcal{P}(d)$ and $H(d)$ for each disease
- 3: Partition diseases into $\mathcal{D}_{\text{unann}}$ and $\mathcal{D}_{\text{ann}}$
- 4: **for all** $d_u \in \mathcal{D}_{\text{unann}}$ **do**
- 5: Find $\mathcal{M}(d_u) = \{d_a \in \mathcal{D}_{\text{ann}} : \mathcal{P}(d_u) = \mathcal{P}(d_a)\}$
- 6: Collect $H_{\text{cand}}(d_u) = \bigcup_{d_a \in \mathcal{M}(d_u)} H(d_a)$
- 7: **for all** $h \in H_{\text{cand}}(d_u)$ **do**
- 8: Compute $s_h(d_u)$
- 9: **if** $s_h(d_u) = 1$ **then**
- 10: Classify h as single-match
- 11: **else if** $s_h(d_u) \geq 2$ **and**
- 12: $s_h(d_u) = \max_{q \in H_{\text{rec}}(d_u)} s_q(d_u)$ **then**
- 13: **if** $s_h(d_u) = |\mathcal{M}(d_u)|$ **then**
- 14: Classify h as fully supported
- 15: **else**
- 16: Classify h as dominant
- 17: **end if**
- 18: **else**
- 19: Classify h as recurrent
- 20: **end if**
- 21: **end for**
- 22: **end for**
- 23: Export matched diseases, candidate terms, support counts, and categories

For reporting, each term is assigned to one non-overlapping category: fully supported, dominant, recurrent, or single-match, in this order. Table VI summarizes these categories.

The exact-matching and HPO-prioritization workflow was

implemented in Python using the Pandas library. The source code and the scripts used to generate the reported outputs are available in the accompanying GitHub repository.

6) *Maude Consistency Verification*: For selected exact-matching disease pairs, the shared Rhea reaction was encoded as a reaction-specific DEA-RS model. Each model contains entities representing a substrate S_{sub} , a product P_{prod} , a cofactor Cof, an enzyme activity state E , and, when applicable, reaction-specific biochemical markers. The model tracks substrate accumulation $S_{\text{sub,acc}}$ and reduced product formation $P_{\text{prod,low}}$ under normal, hypomorphic, null, and cofactor-limited conditions. The three enzyme activity states are E_{norm} , E_{hypo} , and E_{null} . Each scenario is initialized using the following input process:

$$\mathcal{I}_s = \{S_{\text{sub}}\} \cdot \mathbf{0} \parallel \text{rec } X.\text{Env}_s.X.$$

based on the environments:

$$\begin{aligned} \text{Env}_{\text{norm}} &= \{\text{Cof}, E_{\text{norm}}\}, & \text{Env}_{\text{hypo}} &= \{\text{Cof}, E_{\text{hypo}}\}, \\ \text{Env}_{\text{null}} &= \{\text{Cof}, E_{\text{null}}\}, & \text{Env}_{\text{cof-low}} &= \{\text{Cof}_{\text{low}}, E_{\text{norm}}\}. \end{aligned}$$

Reachability queries follow the scheme introduced in Stage 1. They test whether substrate accumulation or reduced product formation becomes reachable under the corresponding perturbation and whether the normal product is no longer reachable. The reaction definitions, DEA-RS rules, and scenario environments for the additional Stage 2 reaction-specific models are reported in Appendix Tables XVI–XVIII.

7) *Mechanistic Consistency Classification*: Each selected candidate association (d_u, h) analyzed in Maude is assigned a mechanistic-consistency class

$$\kappa(d_u, h) \in \{\text{strong, moderate, context-dependent, low}\}.$$

The class describes the relationship between the reaction-level outcome verified in Maude and the candidate HPO term: strong indicates a direct biochemical link, moderate an indirect or downstream link, context-dependent that additional biological or clinical information is required, and low that there is no clear mechanistic link.

Candidate interpretation considers both the support count $s_h(d_u)$ and the mechanistic-consistency class $\kappa(d_u, h)$, as summarized in Table V.

V. RESULTS

The two-stage workflow first validated the Maude encoding against the controlled PT-RS baseline and then applied the resulting framework to rare-disease reaction profiles.

A. Stage 1: Validation Against the PT-RS Baseline

The 38-rule Maude model was evaluated using the 11 reachability properties defined in Table IV. The validation set comprised five negative properties, testing the absence of pathological or incompatible states, and six positive properties, testing the reachability of expected physiological or perturbation-specific outcomes. All 11 properties yielded the expected outcomes, as summarized in Table VII.

Under normal conditions, the negative searches excluded the prohibited states. Incompatible activity states of the same

enzyme were not reachable (P1–P2), and Phe-acc was not reachable when functional PAH was present (P3). Neither QA nor Hawkinsin-acc was reachable under normal HPD activity (P6). No reachable normal state contained both Phe-acc and Tyr-acc (P9).

The positive searches reproduced the expected pathway outcomes. Under PAH-null, a state containing both Phe-acc and PPy was reachable (P4). Additional reachability searches showed that PPy and Glu were reachable in the same scenario, confirming activation of the ArAT-mediated transamination shunt (P10).

Under HPD-hypo, both QA and Hawkinsin-acc were reachable (P5). In both HGD-deficient scenarios, HGA-acc and BQA were reachable (P7).

In the FAH-hypo and FAH-null scenarios, all five markers of the encoded toxic cascade were reachable: FAA-acc, SAA, SA, ALA-acc, and PBG-low (P8). Additional searches showed that AcAc-low and Fum-low were also reachable under FAH-hypo, consistent with reduced formation of the terminal products. Under normal conditions, the physiological terminal products AcAc and Fum were reachable (P11).

The two queries presented in the Methods illustrate a positive and a negative reachability result. For P7, the search under reduced HGD activity returned a state containing HGA-acc. The relevant portion of the returned state was:

```
Maude> search [1] init(input-process-HGD-hypo) =>*
      STATE such that STATE |= (HGA-acc) .

Solution 1 (state 4)
STATE --> {proc: rec 'X . . . . 'X ;
           in: HGA, CO2, HPD-Fe2 ;
           out: HGA-acc, MAA-low, H, HGD-hypo ;
           ctx: DHPR, GSH, H2O, O2, NADH, H, PLP ...
           }
```

The presence of HGA-acc in the out component confirms that HGA accumulation is reachable under HGD-hypo.

For P6, the search for Hawkinsin-acc under normal conditions returned No solution:

```
Maude> search [1] init(input-process-normal) =>*
      STATE such that STATE |= (Hawkinsin-acc) .

No solution.
```

This result confirms that Hawkinsin-acc is not reachable under normal conditions. Overall, the Maude model reproduced the expected physiological and perturbed pathway behavior.

B. Stage 2: Rare-Disease Dataset and Candidate HPO Prioritization

1) *Dataset Composition*: The integration of Orphanet, UniProt, Rhea, and ChEBI yielded a dataset comprising 1,276 rare diseases, 926 enzyme-coding genes, 1,696 reactions, and 6,915 disease-reaction profiles. After applying the cofactor/-pathway filter and retaining only complete six-field profiles, 117 diseases remained in the analysis. Of these, 32 had no recorded HPO annotations, whereas 85 had at least one HPO annotation.

Exact six-field profile matching found at least one HPO-annotated match for 10 of the 32 HPO-unannotated diseases.

TABLE VII: PT-RS validation properties and Maude outcomes.

ID	Validation class	Scenario	Target outcome	Query type	Maude outcome
P1	Enzyme activity–state exclusivity	Normal context	PAH-null is unreachable and does not co-occur with PAH-Fe2.	Negative	All prohibited states unreachable ✓
P2	Enzyme activity–state exclusivity	Normal context	No incompatible activity states of the same enzyme are reachable simultaneously.	Negative	No incompatible pair reachable ✓
P3	Physiological integrity	Normal PAH context	Phe-acc is unreachable.	Negative	No solution ✓
P4	Disease-specific marker reachability	PAH-null context	Phe-acc and PPy are reachable in the same state.	Positive	A joint state containing Phe-acc and PPy is reachable ✓
P5	Disease-specific marker reachability	HPD-hypo	QA and Hawkinsin-acc are reachable.	Positive	Both markers reachable ✓
P6	Physiological integrity	Normal HPD context	QA and Hawkinsin-acc are unreachable.	Negative	Both markers unreachable ✓
P7	Disease-specific marker reachability	HGD-hypo and HGD-null	HGA-acc and BQA are reachable.	Positive	Both markers reachable in both scenarios ✓
P8	Disease-specific marker reachability	FAH-hypo and FAH-null	FAA-acc, SAA, SA, ALA-acc, and PBG-low are reachable.	Positive	All five markers reachable in both scenarios ✓
P9	Physiological integrity	Normal context	Phe-acc and Tyr-acc are not reachable simultaneously.	Negative	No joint state reachable ✓
P10	Pathway-level outcome reachability	PAH-null	PPy and Glu are reachable through the ArAT-mediated transamination shunt.	Positive	Both shunt products reachable ✓
P11	Pathway-level outcome reachability	Normal context	The terminal products AcAc and Fum are reachable.	Positive	Both terminal products reachable ✓

Note: Multi-marker properties were evaluated either through separate searches for each marker or, when simultaneous presence was part of the property, through a conjunctive condition in a single search. Properties involving two perturbation scenarios were tested separately in each scenario. The table summarizes the corresponding results for each property.

TABLE VIII: Distribution of candidate HPO terms by support category.

Candidate category	Support condition	HPO-unannotated diseases	Unique HPO terms	Candidate associations
Single-match (H_{single})	Supported by one exact-matching HPO-annotated disease.	10	391	478
Recurrent (H_{rec})	Supported by at least two exact-matching diseases, but not dominant.	1	16	16
Dominant (H_{dom})	Recurrent term with the highest support.	1	1	1
Fully supported (H_{full})	Present in every exact-matching HPO-annotated disease.	0	0	0
Total	Candidate terms obtained from exact profile matches.	10 (31.3%)	401	495

Note: Counts should not be summed across support categories because the same disease or HPO term may appear in more than one category. The 22 diseases without an exact match are not included because they generated no candidate HPO associations.

TABLE IX: Dataset composition and exact-matching results in Stage 2.

Parameter	Value	Description
Disease–reaction rows	134,209	Input disease–reaction records.
Rows after cofactor/pathway filtering	10,609	Records with at least one cofactor and one pathway.
Diseases with complete profiles	117	Diseases with complete Gene, Entry, EC, Rhea ID, Cofactor, and Pathway fields.
HPO-unannotated diseases	32	Diseases with $H(d) = \emptyset$.
HPO-annotated diseases	85	Diseases with $ H(d) > 0$.
Diseases with an exact match	10	HPO-unannotated diseases with at least one exact-matching HPO-annotated disease.
Diseases without an exact match	22	HPO-unannotated diseases without an identical six-field profile.

The remaining 22 diseases had no exact match. Table IX summarizes the dataset composition and matching results.

Removing the cofactor/pathway filter did not affect the matching results. Because complete six-field profiles were required in both settings, each retained the same 117 diseases: 32 HPO-unannotated and 85 HPO-annotated. Both settings also yielded 10 matched diseases, 22 unmatched diseases, and 495 candidate disease–HPO associations. The full comparison is reported in Appendix Table ??.

2) Exact Profile Matching and Candidate HPO Support:

Exact profile matching identified candidate HPO terms for 10 of the 32 HPO-unannotated diseases. The remaining 22

diseases had no exact match with an HPO-annotated disease.

Across the 10 matched diseases, the analysis identified 495 candidate disease–HPO associations, representing 401 unique HPO terms. Of these, 478 associations were supported by one exact-matching HPO-annotated disease and were classified as *single-match*. ORPHA:90625 was the only disease with multiple exact matches, with one dominant candidate and 16 recurrent candidates. No fully supported candidate was identified (Table VIII).

Table X reports the 10 HPO-unannotated diseases, their exact-matching HPO-annotated diseases, and selected candidate HPO terms. Two rows are highlighted as representative examples.

Tyrosinemia type 3 (ORPHA:69723) matched Hawkinsinuria (ORPHA:2118) through their shared *HPD* profile. The selected single-match candidates were 4-hydroxyphenylpyruvic aciduria (HP:0003161) and metabolic acidosis (HP:0001942).

ORPHA:90625 matched five *PRPS1*-related diseases. Peripheral neuropathy (HP:0009830) was the only dominant candidate in the analysis and was present in four of the five exact-matching diseases. The same disease also had 16 recurrent candidates, listed in Appendix Table ??. The other nine matched diseases had only single-match candidates.

3) *Internal Baseline Comparison:* Exact six-field matching identified 10 of the 32 HPO-unannotated diseases, with an average of 49.5 candidate HPO associations per matched

TABLE X: Exact profile matches and selected candidate HPO terms.

ORPHA	HPO-unannotated disease	Gene(s)	Exact matches (n)	Exact-matching HPO-annotated disease(s)	Support category	Selected candidate HPO term(s)
247198	Progressive cerebello-cerebral atrophy	<i>SEPSECS</i>	1	Pontocerebellar hypoplasia type 2 (2524)	Single-match	HP:0001321 Cerebellar hypoplasia
254905	Isolated cytochrome C oxidase deficiency	<i>MT-CO1</i> <i>MT-CO2</i> <i>MT-CO3</i>	1	MELAS (550)	Single-match	HP:0001638 Cardiomyopathy
293284	Tetrahydrobiopterin-responsive phenylketonuria	<i>PAH</i>	1	Maternal phenylketonuria syndrome (2209)	Single-match	HP:0000752 Hyperactivity
369920	Pontocerebellar hypoplasia type 9	<i>AMPD2</i>	1	Autosomal recessive spastic paraplegia type 63 (401805)	Single-match	HP:0001257 Spasticity
583602	Neu-Laxova syndrome due to phosphoserine aminotransferase deficiency	<i>PSAT1</i>	1	Phosphoserine aminotransferase deficiency, infantile/juvenile form (284417)	Single-match	HP:0001339 Lissencephaly; HP:0012279 Hyposerinemia
642099	Spondyloepimetaphyseal dysplasia with joint laxity, Beighton type	<i>B3GALT6</i>	1	B3GALT6-related spondylodysplastic EDS (536467)	Single-match	HP:0002650 Scoliosis; HP:0002828 Multiple joint contractures
647772	Isolated primary pigmented nodular adrenocortical disease	<i>PDE11A</i>	1	Carney complex (1359)	Single-match	HP:0011672 Cardiac myxoma; HP:0000845 Elevated GH
69723	Tyrosinemia type 3	<i>HPD</i>	1	Hawkinsinuria (2118)	Single-match	HP:0003161 4-Hydroxyphenylpyruvic aciduria; HP:0001942 Metabolic acidosis
708895	Tetrahydrobiopterin-unresponsive phenylketonuria	<i>PAH</i>	1	Maternal phenylketonuria syndrome (2209)	Single-match	HP:0001249 Intellectual disability; HP:0000252 Microcephaly
90625	Rare X-linked non-syndromic sensorineural deafness type DFN	<i>PRPS1</i>	5	Five <i>PRPS1</i> -related diseases (ORPHA:1187, 411536, 411543, 423479, and 99014)	Dominant	HP:0009830 Peripheral neuropathy, present in 4 of 5 exact matches

Key: *n* is the number of exact-matching HPO-annotated diseases, not the number of genes or reaction descriptors. Lilac rows indicate the examples discussed in the text. EDS = Ehlers–Danlos syndrome; GH = growth hormone.

TABLE XI: Internal comparison of matching criteria.

Criterion	Matching information	Matched diseases	Mean HPO associations per matched disease
Gene-only	Causal gene	10/32	49.5
UniProt Entry-only	UniProt entry	10/32	49.5
EC-only	Enzyme class	13/32	77.2
Rhea-only	Reaction	13/32	77.2
Pathway-only	Pathway context	13/32	85.5
Gene+EC+Rhea	Core enzymatic identity	10/32	49.5
Gene+Entry+EC+Rhea	Core protein-reaction profile	10/32	49.5
Exact six-field	Complete biochemical profile	10/32	49.5

Key: Matched diseases are reported out of the 32 HPO-unannotated diseases. For each criterion, the mean is the total number of candidate HPO associations divided by the number of matched diseases.

disease. Gene-only, UniProt Entry-only, Gene+EC+Rhea, and Gene+Entry+EC+Rhea matching gave the same results.

EC-only and Rhea-only matching each identified 13 diseases, with an average of 77.2 candidate HPO associations per matched disease. Pathway-only matching identified the same number of diseases but produced the largest candidate lists, with an average of 85.5 associations per matched disease (Table XI).

Exact six-field matching was used in the subsequent analysis because it preserves all six biochemical fields required for the reaction-level Maude models.

4) *Maude Consistency Verification*: Selected exact-matching disease pairs were analyzed in Maude to determine whether perturbation of their shared reaction made the expected biochemical markers reachable. The PAH and HPD components of the Stage 1 Phe–Tyr model were reused for the phenylketonuria and tyrosinemia type 3 comparisons. Four additional reaction-specific DEA–RS models were constructed for AMPD2, PSAT1, SEPSECS, and B3GALT6 (see Appendix Tables XVI–XVIII for the complete model specifications).

a) *Reaction-specific models*: AMPD2 catalyses the Zn^{2+} -dependent conversion of AMP to IMP and NH_4^+ in purine metabolism. PSAT1 catalyses a PLP-dependent step in L-serine biosynthesis that forms phosphoserine. SEPSECS participates in the PLP-dependent conversion of O-phosphoseryl-tRNA(Sec) to selenocysteinyl-tRNA(Sec). B3GALT6 is a Mn^{2+} -dependent galactosyltransferase that extends the proteoglycan-linker structure. These reactions were modeled separately using their corresponding substrates, products, and cofactors.

Each model was tested under normal, hypomorphic, null, or cofactor-limited conditions, as appropriate. The queries assessed substrate accumulation, reduced product formation, and absence of the normal product. The reachable markers were then compared with the candidate HPO terms to assign the mechanistic-consistency class κ (Table XII).

For example, the PSAT1 comparison illustrates how the table should be read. Under PSAT1 perturbation, P-OxoPyr-acc and P-Ser-low were reachable, indicating substrate accumulation and reduced phosphoserine formation. This provides a direct biochemical basis for hyposerinemia (HP:0012279), and the association was therefore classified as *strong*. The corresponding row is highlighted in Table XII.

b) *Effects of enzyme and cofactor perturbations*: In the AMPD2 model, Zn^{2+} limitation made AMP-acc and IMP-low reachable, reproducing the qualitative effect of reduced enzyme activity. Under null activity, normal IMP formation was not reachable. PLP limitation produced the same reduced-activity pattern in the PSAT1 and SEPSECS models, whereas Mn^{2+} limitation had the same effect in the B3GALT6 model. In all four models, null activity prevented formation of the normal product.

All selected models returned the expected reaction-level markers. Direct biochemical links were classified as *strong*,

TABLE XII: Reaction-level Maude results and mechanistic-consistency classes for selected exact-matching disease pairs.

Gene / Rhea ID	HPO-unannotated disease	Exact-matching HPO-annotated disease	Support category	Candidate HPO term	Reachable marker / κ
HPD RHEA:16189	Tyrosinemia type 3 (69723)	Hawkinsinuria (2118)	Single-match	HP:0003161 4-Hydroxyphenylpyruvic aciduria	HPP-acc; direct biochemical link; <i>strong</i>
HPD RHEA:16189	Tyrosinemia type 3 (69723)	Hawkinsinuria (2118)	Single-match	HP:0001942 Metabolic acidosis	HPP-acc; downstream systemic link; <i>moderate</i>
PAH RHEA:20273	Tetrahydrobiopterin-responsive phenylketonuria (293284)	Maternal phenylketonuria syndrome (2209)	Single-match	HP:0000752 Hyperactivity	Phe-acc and PPy; downstream link; <i>moderate</i>
PAH RHEA:20273	Tetrahydrobiopterin-unresponsive phenylketonuria (708895)	Maternal phenylketonuria syndrome (2209)	Single-match	HP:0001249 Intellectual disability; HP:0000252 Microcephaly	Phe-acc and PPy; downstream links; <i>moderate</i>
SEPSECS RHEA:25041	Progressive cerebello-cerebral atrophy (247198)	Pontocerebellar hypoplasia type 2 (2524)	Single-match	HP:0001321 Cerebellar hypoplasia	Ser-tRNA-Sec-acc and Sec-tRNA-Sec-low; additional developmental context required; <i>context-dependent</i>
AMPD2 RHEA:14777	Pontocerebellar hypoplasia type 9 (369920)	Autosomal recessive spastic paraplegia type 63 (401805)	Single-match	HP:0001257 Spasticity	AMP-acc and IMP-low; downstream neurological link; <i>moderate</i>
PSAT1 RHEA:14329	Neu-Laxova syndrome due to phosphoserine aminotransferase deficiency (583602)	Phosphoserine aminotransferase deficiency, infantile/juvenile form (284417)	Single-match	HP:0012279 Hyposerinemia	P-OxoPyr-acc and P-Ser-low; direct biochemical link; <i>strong</i>
PSAT1 RHEA:14329	Neu-Laxova syndrome due to phosphoserine aminotransferase deficiency (583602)	Phosphoserine aminotransferase deficiency, infantile/juvenile form (284417)	Single-match	HP:0001339 Lissencephaly	P-OxoPyr-acc and P-Ser-low; downstream neurodevelopmental link; <i>moderate</i>
B3GALT6 RHEA:11780	Spondyloepimetaphyseal dysplasia with joint laxity, Beighton type (642099)	B3GALT6-related spondylodysplastic EDS (536467)	Single-match	HP:0002650 Scoliosis; HP:0002828 Multiple joint contractures	Prot-Xyl-Gal-acc and Prot-Xyl-Gal-Gal-low; downstream skeletal links; <i>moderate</i>

Key: κ denotes the mechanistic-consistency class relating the reaction-level marker to the candidate HPO term. Rhea IDs identify the biochemical reactions represented in the models. The lilac row is the example described in the text.

Gene symbols: HPD = 4-hydroxyphenylpyruvate dioxygenase; PAH = phenylalanine hydroxylase; SEPSECS = O-phosphoserine:selenocysteinyl-tRNA synthase; AMPD2 = adenosine monophosphate deaminase 2; PSAT1 = phosphoserine aminotransferase 1; B3GALT6 = beta-1,3-galactosyltransferase 6.

Model notation: acc = accumulation; low = reduced product formation; HPP = 4-hydroxyphenylpyruvate; Phe = phenylalanine; PPy = phenylpyruvate; AMP = adenosine monophosphate; IMP = inosine monophosphate; P-OxoPyr = phosphohydroxypyruvate; P-Ser = phosphoserine. Ser-tRNA-Sec and Sec-tRNA-Sec denote the substrate and product states represented in the SEPSECS model, respectively. Prot-Xyl-Gal and Prot-Xyl-Gal-Gal denote proteoglycan-linker intermediates containing one and two galactose residues, respectively. EDS = Ehlers-Danlos syndrome.

whereas candidate terms related to downstream clinical effects were classified as *moderate*. The SEPSECS association with cerebellar hypoplasia was classified as *context-dependent*, because the reachable tRNA-level defect alone did not explain the developmental phenotype.

These findings provide reaction-level support for the selected candidate associations rather than direct predictions of clinical phenotypes.

VI. DISCUSSION

A. Main Findings and Candidate Support

Candidate HPO terms were considered only after an HPO-unannotated disease matched an HPO-annotated disease on all six biochemical fields, so prioritization was based on shared biochemistry rather than phenotypic similarity. Stage 1 reproduced the expected Phe-Tyr pathway behavior and supported the use of the same DEA-RS logic in Stage 2.

Exact six-field matching identified candidate HPO terms for 10 of the 32 HPO-unannotated diseases. Most were single-match candidates, reflecting the rarity of identical complete profiles. Despite the lack of recurrence, these associations remained biochemically traceable. For example, tetrahydrobiopterin-responsive phenylketonuria (ORPHA:293284) and tetrahydrobiopterin-unresponsive phenylketonuria (ORPHA:708895) matched maternal phenylketonuria

syndrome (ORPHA:2209) through their shared PAH profiles. Tyrosinemia type 3 (ORPHA:69723) matched Hawkinsinuria (ORPHA:2118) through the shared HPD profile. These candidates should be considered testable hypotheses for expert review.

ORPHA:90625 provided the clearest recurrence signal. **Peripheral neuropathy (HP:0009830)** was present in four of its five exact matches and was the only dominant candidate in the analysis. This shows how recurrence across diseases with the same complete profile can strengthen candidate prioritization.

EC-only, Rhea-only, and Pathway-only matching each identified three additional diseases but produced longer candidate lists. Pathway-only matching produced the largest lists, showing that pathway information alone was the least selective. Exact six-field matching was chosen because it was more selective and preserved the information required for Maude analysis.

For the selected candidates examined in Maude, interpretation considered both recurrence among exact matches and the directness of the link to the reachable biochemical marker.

B. Role of Maude Verification

Maude does not predict HPO terms directly. It tests whether biochemical markers, such as substrate accumulation or reduced product formation, are reachable under enzyme or cofactor

perturbation. These markers were then compared with the candidate HPO terms reported in Table XII.

The table illustrates three of the four consistency classes. In the HPD model, HPP-acc directly reflects 4-hydroxyphenylpyruvic aciduria (HP:0003161). Similarly, in the PSAT1 model, P-Ser-low provides a direct biochemical basis for hyposerinemia (HP:0012279). These associations were classified as *strong*.

Other candidates, including metabolic acidosis, hyperactivity, spasticity, and lissencephaly, were compatible with the reachable reaction-level markers but represented downstream clinical effects. They were therefore classified as *moderate*. The SEPSECS association with cerebellar hypoplasia was classified as *context-dependent*, because the reachable Ser-tRNA-Sec-acc and Sec-tRNA-Sec-low markers did not alone explain the developmental phenotype.

The AMPD2, PSAT1, SEPSECS, and B3GALT6 models also showed that cofactor limitation reproduced the qualitative effect of reduced enzyme activity, whereas null activity prevented normal product formation. Maude reachability analysis thus provides a transparent reaction-level check that helps distinguish direct biochemical links from indirect phenotype associations without confirming the clinical phenotype itself.

VII. CONCLUSION

This work shows how reaction-level information can help prioritize candidate HPO terms in rare enzymatic diseases. Biochemical matching is performed before HPO terms are considered, so phenotypic similarity does not influence the disease match. A candidate term is proposed only when an HPO-unannotated disease shares the same six-field biochemical profile with an HPO-annotated disease.

Maude reachability analysis adds a second level of evaluation. It does not predict the clinical phenotype, but tests whether the corresponding enzyme or cofactor perturbation produces the expected biochemical effect. This helps distinguish candidates with a direct biochemical link from those related to downstream or context-dependent effects.

The main limitation is coverage. Exact six-field matching leaves many HPO-unannotated diseases unmatched, although it keeps each candidate traceable to a defined gene, protein, enzyme class, reaction, cofactor, and pathway. In addition, the one-reaction models used in Stage 2 cannot represent multi-step or cross-pathway mechanisms. The framework should therefore support expert review rather than serve as an automatic HPO annotation system.

Future work will examine controlled partial-profile matching, more complex reaction models, and validation against diseases with curated HPO annotations. These extensions may increase coverage while preserving the biochemical traceability and transparent reasoning provided by the current framework.

The implementation, Maude specifications, and complete output files are available in the project repository [46].

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APPENDIX A ADDITIONAL STAGE 2 ANALYSES

This appendix reports additional analyses and results supporting the main Results section. Full output files and Maude source files are available in the [project repository](#).

A. Unmatched target diseases

Table XIII lists the 22 HPO-unannotated targets that had no exact reaction-profile match with an annotated donor.

B. Filter-sensitivity analysis

Removing the cofactor/pathway filter did not change the exact matching output once complete six-field profiles were still required (Table XIV).

C. Dominant and recurrent HPO candidates for ORPHA:90625

ORPHA:90625 was the only target with multi-donor HPO support. Table XV reports its dominant and recurrent candidate terms.

D. Reaction-specific Maude models for selected disease pairs

The additional reaction-specific DEA-RS models used in the Stage 2 Maude consistency analysis are documented in Tables XVI–XVIII. Table XVI reports the reference biochemical reactions, Table XVII gives the corresponding DEA-RS reaction rules, and Table XVIII summarizes the physiological and perturbed scenario environments.

TABLE XIV: Filter-sensitivity analysis.

Metric	ON	OFF	Δ
Disease pool before complete-profile check	117	1,276	–
Diseases with complete profiles	117	117	0
Targets	32	32	0
Donors	85	85	0
Matched targets	10	10	0
No-match targets	22	22	0
Target–HPO associations	495	495	0

Key: ON applies the cofactor/pathway filter; OFF removes it. Both settings still require complete six-field profiles: Gene, Entry, EC, Rhea_ID, Cofactor, and Pathway.

TABLE XV: Dominant and recurrent HPO candidates for ORPHA:90625.

Candidate HPO term	Support
HP:0009830 Peripheral neuropathy	D; 4/5
HP:0000407 Sensorineural hearing impairment	R; 3/5
HP:0001251 Ataxia	R; 3/5
HP:0000083 Renal insufficiency	R; 2/5
HP:0000648 Optic atrophy	R; 2/5
HP:0000707 Abnormality of the nervous system	R; 2/5
HP:0000791 Uric acid nephrolithiasis	R; 2/5
HP:0001249 Intellectual disability	R; 2/5
HP:0001284 Areflexia	R; 2/5
HP:0001324 Muscle weakness	R; 2/5
HP:0001369 Arthritis	R; 2/5
HP:0001919 Acute kidney injury	R; 2/5
HP:0002149 Hyperuricemia	R; 2/5
HP:0003149 Hyperuricosuria	R; 2/5
HP:0003240 Increased phosphoribosylpyrophosphate synthetase activity	R; 2/5
HP:0008936 Axial hypotonia	R; 2/5
HP:0020074 Crystalluria	R; 2/5

Key: ORPHA:90625 = rare X-linked non-syndromic sensorineural deafness type DFN. D = dominant candidate; R = recurrent candidate. Donor count is calculated over five exact donors.

TABLE XIII: HPO-unannotated targets with no exact reaction-profile match.

ORPHA	Target disease	ORPHA	Target disease
1561	Fatal infantile cytochrome C oxidase deficiency	168598	Methionine adenosyltransferase I/III deficiency
199285	Hereditary hypercarotenemia and vitamin A deficiency	280586	Chondrodysplasia with joint dislocations, gPAPP type
284139	Larsen-like syndrome, B3GAT3 type	289290	Hypermethioninemia encephalopathy due to adenosine kinase deficiency
313850	Infantile cerebellar-retinal degeneration	319678	Encephalopathy-hypertrophic cardiomyopathy-renal tubular disease syndrome
394529	Multiple acyl-CoA dehydrogenase deficiency, severe neonatal type	394532	Multiple acyl-CoA dehydrogenase deficiency, mild type
401859	Lipoic acid synthetase deficiency	457185	Neonatal encephalomyopathy-cardiomyopathy-respiratory distress syndrome
521438	Congenital vertebral-cardiac-renal anomalies syndrome	528084	Non-specific syndromic intellectual disability
597733	Oculocutaneous albinism type 8	647782	Isolated micronodular adrenocortical disease
658778	COQ7-related distal hereditary motor neuropathy	90641	Rare mitochondrial non-syndromic sensorineural deafness
93602	Xanthinuria type II	98676	Autosomal recessive isolated optic atrophy
99731	Isolated sulfite oxidase deficiency	99849	Glycogen storage disease due to muscle beta-enolase deficiency

TABLE XVI: Reference enzymatic reactions used in the additional Stage 2 reaction-specific Maude models.

Rhea ID	Biochemical reaction	EC	Enzyme name	Cofactor
RHEA:25041	O-phospho-L-seryl-tRNA(Sec) + selenophosphate + H ₂ O → L-selenocysteinyl-tRNA(Sec) + 2 phosphate	2.9.1.2	SEPSECS: O-phospho-L-seryl-tRNA(Sec):L-selenocysteinyl-tRNA synthase	PLP
RHEA:14777	AMP + H ₂ O + H ⁺ → IMP + NH ₄ ⁺	3.5.4.6	AMPD2: AMP deaminase	Zn ²⁺
RHEA:14329	O-phospho-L-serine + 2-oxoglutarate → 3-phosphooxypyruvate + L-glutamate	2.6.1.52	PSAT1: phosphoserine aminotransferase	PLP
RHEA:11780	3-O-(β-D-galactosyl-(1→4)-β-D-xylosyl)-L-seryl-[protein] + UDP-α-D-galactose → 3-O-(β-D-galactosyl-(1→3)-β-D-galactosyl-(1→4)-β-D-xylosyl)-L-seryl-[protein] + UDP + H ⁺	2.4.1.134	B3GALT6: galactosylxylosylprotein 3-β-galactosyltransferase	Mn ²⁺

Key: Rhea ID = identifier in the Rhea reaction database; EC = Enzyme Commission number.

Enzymes: SEPSECS = O-phospho-L-seryl-tRNA(Sec):L-selenocysteinyl-tRNA synthase; AMPD2 = AMP deaminase; PSAT1 = phosphoserine aminotransferase; B3GALT6 = galactosylxylosylprotein 3-β-galactosyltransferase.

Metabolites and cofactors: AMP = adenosine monophosphate; IMP = inosine monophosphate; PLP = pyridoxal 5'-phosphate; UDP = uridine diphosphate; tRNA(Sec) = selenocysteine-specific transfer RNA; Zn²⁺ = zinc ion; Mn²⁺ = manganese ion.

TABLE XVII: DEA-RS reaction rules used in the additional Stage 2 reaction-specific Maude models.

ID	Step / condition	Reactants, <i>R</i>	Inhibitors, <i>I</i>	Products, <i>P</i>
M1	PSAT normal activity	P-OxoPyr, Glu, PLP, PSAT	PSAT-hypo, PSAT-null	P-Ser, alpha-KG, PLP, PSAT
M2	PSAT partial activity	P-OxoPyr, Glu, PLP, PSAT-hypo	PSAT	P-Ser-low, P-OxoPyr-acc, Glu, PLP, PSAT-hypo
M3	PSAT null activity	P-OxoPyr, Glu, PLP, PSAT-null	PSAT	P-OxoPyr-acc, Glu, PLP, PSAT-null
M4	PSAT PLP limitation	P-OxoPyr, Glu, PLP-low, PSAT	PLP	P-Ser-low, P-OxoPyr-acc, Glu, PLP-low, PSAT
M5	AMP-deaminase normal activity	AMP, H ₂ O, H, Zn, AMP-deaminase	AMP-deaminase-hypo, AMP-deaminase-null	IMP, NH ₄ , Zn, AMP-deaminase
M6	AMP-deaminase partial activity	AMP, H ₂ O, H, Zn, AMP-deaminase-hypo	AMP-deaminase	IMP-low, NH ₄ , AMP-acc, Zn, AMP-deaminase-hypo
M7	AMP-deaminase null activity	AMP, H ₂ O, H, AMP-deaminase-null	AMP-deaminase	AMP-acc, H ₂ O, H, AMP-deaminase-null
M8	AMP-deaminase Zn limitation	AMP, H ₂ O, H, Zn-low, AMP-deaminase	Zn	IMP-low, NH ₄ , AMP-acc, Zn-low, AMP-deaminase
M9	Galactosyltransferase normal activity	Prot-Xyl-Gal, UDP-Gal, Mn, galactosyltransferase	galactosyltransferase-hypo, galactosyltransferase-null	Prot-Xyl-Gal-Gal, UDP, H, Mn, galactosyltransferase
M10	Galactosyltransferase partial activity	Prot-Xyl-Gal, UDP-Gal, Mn, galactosyltransferase-hypo	galactosyltransferase	Prot-Xyl-Gal-Gal-low, Prot-Xyl-Gal-acc, UDP, Mn, H, galactosyltransferase-hypo
M11	Galactosyltransferase null activity	Prot-Xyl-Gal, UDP-Gal, galactosyltransferase-null	galactosyltransferase	Prot-Xyl-Gal-acc, UDP-Gal, galactosyltransferase-null
M12	Galactosyltransferase Mn limitation	Prot-Xyl-Gal, UDP-Gal, Mn-low, galactosyltransferase	Mn	Prot-Xyl-Gal-Gal-low, Prot-Xyl-Gal-acc, UDP, Mn-low, H, galactosyltransferase
M13	SEPSECS normal activity	Ser-tRNA-Sec, selenophosphate, H ₂ O, PLP, SEPSECS	SEPSECS-hypo, SEPSECS-null	Ser-tRNA-Sec, phosphate, phosphate, PLP, SEPSECS
M14	SEPSECS partial activity	Ser-tRNA-Sec, selenophosphate, H ₂ O, PLP, SEPSECS-hypo	SEPSECS	Ser-tRNA-Sec-low, Ser-tRNA-Sec-acc, phosphate, PLP, SEPSECS-hypo
M15	SEPSECS null activity	Ser-tRNA-Sec, selenophosphate, SEPSECS-null	SEPSECS	Ser-tRNA-Sec-acc, selenophosphate, SEPSECS-null
M16	SEPSECS PLP limitation	Ser-tRNA-Sec, selenophosphate, H ₂ O, PLP-low, SEPSECS	PLP	Ser-tRNA-Sec-low, Ser-tRNA-Sec-acc, phosphate, PLP-low, SEPSECS

TABLE XVIII: Maude environments used to construct the physiological and perturbed scenarios for the additional Stage 2 reaction-specific models.

Scenario and environment	Environment entities	Biological interpretation
SEPSECS normal physiological scenario env-normal	env-base,PLP,SEPSECS	Reference SEPSECS scenario with functional enzyme and adequate PLP availability.
SEPSECS partial deficiency env-SEPSECS-hypo	env-base,PLP,SEPSECS-hypo	Partial SEPSECS-deficiency scenario with reduced selenocysteinyl-tRNA(Sec) formation and accumulation of the upstream substrate.
SEPSECS complete deficiency env-SEPSECS-null	env-base,PLP,SEPSECS-null	SEPSECS-null scenario representing complete loss of functional SEPSECS activity and blockade of selenocysteinyl-tRNA(Sec) formation.
SEPSECS cofactor limitation env-PLP-low	env-base,PLP-low,SEPSECS	PLP-limited scenario in which reduced cofactor availability induces SEPSECS-hypo-like behaviour despite the presence of functional SEPSECS.
Galactosyltransferase normal physiological scenario env-normal	env-base,Mn,galactosyltransferase	Reference glycosylation scenario with functional galactosyltransferase and adequate Mn^{2+} availability.
Galactosyltransferase partial deficiency env-galactosyltransferase-hypo	env-base,Mn,galactosyltransferase-hypo	Partial galactosyltransferase-deficiency scenario with reduced formation of the galactosylated proteoglycan product and precursor accumulation.
Galactosyltransferase complete deficiency env-galactosyltransferase-null	env-base,Mn,galactosyltransferase-null	Galactosyltransferase-null scenario representing complete loss of enzyme activity and blockade of the corresponding glycosylation step.
Galactosyltransferase cofactor limitation env-Mn-low	env-base,Mn-low,galactosyltransferase	Mn^{2+} -limited scenario in which reduced cofactor availability impairs galactosyltransferase activity despite the presence of the functional enzyme.
AMP-deaminase normal physiological scenario env-normal	env-base,Zn,AMP-deaminase	Reference purine-metabolism scenario with functional AMP deaminase and adequate Zn^{2+} availability.
AMP-deaminase partial deficiency env-AMP-deaminase-hypo	env-base,Zn,AMP-deaminase-hypo	Partial AMP-deaminase-deficiency scenario with reduced IMP formation and accumulation of AMP.
AMP-deaminase complete deficiency env-AMP-deaminase-null	env-base,Zn,AMP-deaminase-null	AMP-deaminase-null scenario representing complete loss of enzyme activity and accumulation of AMP.
AMP-deaminase cofactor limitation env-Zn-low	env-base,Zn-low,AMP-deaminase	Zn^{2+} -limited scenario in which reduced zinc availability impairs AMP-deaminase activity and decreases IMP formation.
PSAT normal physiological scenario env-normal	env-base,PLP,PSAT	Reference phosphoserine-aminotransferase scenario with functional PSAT and adequate PLP availability.
PSAT partial deficiency env-PSAT-hypo	env-base,PLP,PSAT-hypo	Partial PSAT-deficiency scenario with reduced O-phospho-L-serine formation and accumulation of 3-phospho-hydroxypyruvate.
PSAT complete deficiency env-PSAT-null	env-base,PLP,PSAT-null	PSAT-null scenario representing complete loss of PSAT activity and blockade of O-phospho-L-serine formation.
PSAT cofactor limitation env-PLP-low	env-base,PLP-low,PSAT	PLP-limited scenario in which reduced cofactor availability induces PSAT-hypo-like behaviour despite the presence of functional PSAT.

Environment templates: for the SEPSECS model, env-base=(H₂O); for the galactosyltransferase model, env-base=(UDP-Gal); for the AMP-deaminase model, env-base=(H₂O,H); for the PSAT model, env-base=(Glu). The suffix -hypo denotes partial enzymatic activity, whereas -null denotes complete loss of enzymatic activity. The suffix -low denotes reduced cofactor availability. PLP = pyridoxal 5'-phosphate; Mn = Mn^{2+} ; Zn = Zn^{2+} .